

■■■ SIMShield

Enterprise Mobile Identity Fraud Defense & Zero-Trust Banking Interception Platform

Core Breakthrough: SIMShield solves telecom-based account takeover (SIM Swaps, eSIM hijacks, SS7 SMS interception) by enforcing **Pre-Transaction Zero-Trust Interception**. Fraudulent transfers are evaluated and blocked **BEFORE** simulated money movement occurs, guaranteeing **Zero Balance Loss** (■■10,000 stays untouched) without relying on compromised SMS OTP channels.

1. Unified Architecture & Server-Authoritative Model

SIMShield operates under a strict **Zero-Trust Client Principle**: the mobile application is an untrusted presentation layer. All risk scores (0–100), mitigation policies (ALLOW, WARN, STEP_UP, BLOCK), and banking state changes are computed solely on the authoritative Node.js backend service.

- **Single Unified Android Application:** Integrates **Payment Gateway** and **Security Defense Center** into a single native Android APK (API 26–35) with smooth tab switching.
- **Pre-Transaction Interception Pipeline:** Mandatory pre-check API evaluates 24-hour temporal signals before allowing ledger debit execution.
- **Dual-Mode Operation:** Seamlessly connects to live backend over Wi-Fi (via in-app ■■■ Server Config) or executes standalone on-device zero-trust simulation when offline.
- **Zero Raw PII Storage:** Plaintext phone numbers and device IDs are never persisted; all identifiers are irreversibly salted with HMAC-SHA256.

2. Mobile Application Features (SIMShield App v1.1.0)

Component	Capabilities & Interactive Features
■■ Payment Gateway (Mock UPI Tab)	• Real-time simulated ledger balance (■■10,000 INR) • Quick contact chips: Rahul (■■2k), Priya (■■1k), Amit (■■500), Attacker Mule (■■2k) • Real-time amount watcher updating interactive pay CTA dynamically • Pre-check interception: normal payments succeed (■■8,000); ATO fraud displays ■ TRANSACTION BLOCKED alert with untouched ■10,000 balance • Full transaction ledger audit history & in-app Developer Sandbox
■■■ Security Center	• Hero 0–100 risk score gauge & live telecom SIM/eSIM threat status • 24-hour sliding temporal correlation feed of security events • 'This was me': Native Android BiometricPrompt (fingerprint/face) out-of-band identity verification • 'Secure My Account': Emergency panic lockdown freezing outbound transfers & revoking sessions
■■■ Header Controls	• ■■■ Server Config Dialog: Connect to physical phone over local Wi-Fi or emulator with persistent preferences • ■ Sync Pill Button: Instant tactile data refresh with toast feedback

3. Multi-Signal Scoring Engine & 24-Hour Temporal Window

Signals from telecom partners, authentication gateways, and banking velocity are correlated within a **24-hour sliding temporal window** and capped at 100 points:

Signal Event	Weight	Domain	Standard Reason Code
SIM_CHANGED / REPLACED	+30	Telecom Physical	RECENT_SIM_CHANGE
ESIM_CHANGED / ADDED	+30	Telecom Digital	RECENT_ESIM_CHANGE
NUMBER_PORTED	+25	Telecom Routing	NUMBER_PORTED
NEW_DEVICE_LOGIN	+20	Authentication	NEW_DEVICE
PASSWORD_RESET	+15	Credential Recovery	PASSWORD_RESET_AFTER_SIM_CHANGE
NEW_BENEFICIARY	+15	Core Banking	NEW_BENEFICIARY_AFTER_SIM_CHANGE
ACCOUNT_TAKEOVER_PATTERN	95	Correlated Multi-Signal	ACCOUNT_TAKEOVER_PATTERN

Automated Policy Thresholds:

- **0 – 29 LOW (ALLOW):** Normal operations. Direct fund execution.
- **30 – 49 MEDIUM (WARN_USER):** Advisory warning banner. In-app biometric confirmation recommended.
- **50 – 79 HIGH (STEP_UP_AUTH):** Hardware BiometricPrompt required before processing sensitive transactions.
- **80 – 100 CRITICAL (HOLD / BLOCK):** Transmit **BLOCK** decision. Preserves balance (₹10,000) with zero loss; files P1 incident.

4. End-to-End Live Hackathon Demonstration Flow

Step 1 (Initial Posture): Open SIMShield app. Balance displays ₹10,000 INR; security posture is **LOW RISK** (18/100).

Step 2 (Positive Control): Send ₹2,000 to Rahul. Pre-check returns ALLOW. Payment succeeds; balance updates to ₹8,000.

Step 3 (Attack Injection): Reset balance to ₹10,000. Tap [SIMULATE FULL ACCOUNT TAKEOVER (ATO)]. Simulated attack chain runs (SIM Swap + New Device + Password Reset + Beneficiary). Risk score jumps to **95/100 (CRITICAL)**.

Step 4 (Hero Interception): Attempt to transfer ₹2,000 to Attacker Mule. Pre-check triggers BLOCK. High-priority alert modal appears stating transaction was blocked before execution. **Balance remains strictly ₹10,000 untouched!**

Step 5 (Panic Lockdown): Tap [**SECURE MY ACCOUNT**]. Account transitions to PROTECTED, freezing all outbound transfers and dispatching a P1 SecOps case.

5. Automated Test Suite & Codebase Invariants

Verification Results (34/34 Automated Tests Passing):

- **SecOps Studio Suite (7 tests):** Dashboard rendering, metrics aggregation, case escalation, state reset.
- **HTTP API Integration Suite (7 tests):** Liveness probes, HMAC event ingestion, user risk queries, fraud reporting.
- **Risk Scoring & Correlation Suite (13 tests):** Temporal 24h window expiration, reason codes, crypto HMAC signatures.
- **Mock UPI & Anti-Fraud Suite (7 tests):** Legitimate debits, biometric step-up, critical ATO block, zero-loss invariants.

SIMShield Security Team — Comprehensive Zero-Trust Mobile Identity & Fraud Defense Platform